

Latent Class Analysis of Substance Use Patterns: A Tutorial Using NSDUH Data

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Introduction

Substance use data is often **categorical** in nature — surveys will record whether a respondent used a given substance in the past year (yes/no), the frequency of use (daily/weekly/monthly/never), and so on. Classic approaches to categorical data analysis (CDA) such as contingency tables, log-linear models, and mosaic displays are tools for exploring pairwise relationships between substances. However, when we have many substances recorded simultaneously, we quickly face the challenge of “high-dimensional” categorical data. For instance, if you have six binary indicators, there are $2^6 = 64$ possible response patterns, some/many of which will be sparsely observed, making traditional contingency table approaches impractical.

Latent Class Analysis (LCA) offers a solution to this problem. It assumes that the observed pattern of substance use across many variables can be explained by membership in a small number of latent or unobserved classes. Each class is a profile of item-response probabilities. For example, one class might have high probability of alcohol use and low probability for all other substances, while another class might show uniformly high probabilities across multiple substances (a “polysubstance” profile).

This tutorial demonstrates LCA using the National Survey on Drug Use and Health (NSDUH; <https://www.samhsa.gov/data/data-we-collect/nsduh-national-survey-drug-use-and-health>), an annual U.S. survey of approximately 70,000 individuals aged 12 and older. In this tutorial, we will achieve several goals:

1. Load and prepare the public use NSDUH file
2. Explore the data with classic CDA tools (e.g., contingency tables, mosaic plots)
3. Fit LCA models with varying numbers of classes using `poLCA`
4. Select the best-fitting model using information criteria
5. Interpret and name the resulting latent classes
6. Profile the classes by demographic characteristics including sex and age

To begin, let's load the required packages

```
install.packages(c("poLCA", "tidyverse", "vcd", "vcdExtra", "ggplot2",  
"kableExtra"))
```

```
library(poLCA)      # latent class analysis
library(tidyverse)  # data wrangling
library(vcd)        # mosaic plots and CDA
library(vcdExtra)   # extended CDA tools
library(ggplot2)    # plotting
library(kableExtra) # formatted tables
```

1. Getting the Data: The Public Use NSDUH 2024 File

Downloading the Data

The most recent Public Use NSDUH File is from 2024 and it is freely available from SAMHSA, the Substance Abuse and Mental Health Services Administration:

1. Go to: <https://www.samhsa.gov/data/data-we-collect/nsduh-national-survey-drug-use-and-health/datafiles>
2. Click on 2024 (or select whichever file you are interested in analyzing, and edit the code below according to the year of the file)
3. Download the corresponding R data file
4. Place the downloaded file in your project's working directory

```
# Loading the NSDUH 2024 R data file
load("NSDUH_2024.RData")
```

Identifying Key Variables

The NSDUH code book (https://www.samhsa.gov/data/system/files/media-puf-file/nsduh-2024-ds0001-info-codebook_v1.pdf) and data set contains thousands of variables. In this tutorial, we are focusing on six binary past-year substance use indicators: alcohol, tobacco/nicotine, marijuana/cannabis, cocaine, prescription pain medication misuse, and hallucinogens. We select these variables and add them to the data frame “var_table”, from which we can make a nice table showing the variables of interest and how they are coded (Table 1).

Table 1. Substance Use Indicator Variables from NSDUH 2024

Variable	Description	Coding
ALCYR	Past year alcohol use	0 = No, 1 = Yes
TOBYR	Past year tobacco/nicotine product use	0 = No, 1 = Yes
MRJYR	Past year marijuana/cannabis use	0 = No, 1 = Yes
COCYR	Past year cocaine use	0 = No, 1 = Yes
PNRNMYR	Past year prescription drug misuse	0 = No, 1 = Yes
HALLUCYR	Past year hallucinogen use	0 = No, 1 = Yes

2. Exploratory Categorical Data Analysis

Before beginning to fit our Latent Class Analysis, we examine (some of) the pairwise associations between substances using classic CDA tools. This section demonstrates the utility and limitations of these approaches when applied to multivariate categorical data, motivating the need for the LCA. For instance, there are 15 possible pairwise associations given the 6 substance use variables, which can quickly get overwhelming (a 6-way contingency table creates $2^6 = 64$ cells). LCA can summarize the associations among all six substances using only a small number of parameters, making it far more reasonable than modelling the 64 cells of the 6-way contingency table. This connection to the broader logic of categorical data analysis, where we seek to explain associations among variables through simpler underlying structure, is what makes LCA a natural extension of the methods covered in the PSYC 6136 course.

First, we produce the marginal frequencies to see how often each substance is endorsed in the data.

Marginal Frequencies

```
substance_names <- c("ALCYR", "TOBYR", "MRJYR", "COCYR", "PNRNMYSR",  
"HALLUCYR")  
  
# Compute prevalence of each substance (proportion using = coded as 1)  
  
prevalence <- NSDUH_2024 |>  
  dplyr::select(all_of(substance_names)) |>  
  dplyr::summarise(across(everything(), list(  
    N = ~ sum(. == 1),  
    Prevalence = ~ mean(. == 1)))) |>  
  pivot_longer(everything(), names_to = c("Substance", ".value"), names_sep  
= "_") |>  
  arrange(desc(Prevalence))  
  
# Build table  
  
prevalence |>  
  mutate(Prevalence = scales::percent(Prevalence, accuracy = 0.1)) |>  
  kable(caption = "Table 2. Past-Year Prevalence of Each Substance. Total  
sample size: N = 58,633",  
        col.names = c("Substance", "N (Yes)", "Prevalence"))
```

Table 2. Past-Year Prevalence of Each Substance. Total sample size: N = 58,633

Substance	N (Yes)	Prevalence
ALCYR	33939	57.9%

Substance	N (Yes)	Prevalence
MRJYR	14589	24.9%
TOBYR	12000	20.5%
HALLUCYR	2584	4.4%
PNRNMYR	1673	2.9%
COCYR	991	1.7%

We then also created a plot to visualize the marginal frequencies of each substance variable:

Build plot

```
ggplot(prevalence, aes(x = reorder(Substance, Prevalence), y = Prevalence)) +
  geom_col(fill = "#2c7bb6", alpha = 0.85) +
  geom_text(aes(label = paste0("n = ", format(N, big.mark = ", "))),
            hjust = -0.1,
            size = 3.5) +
  coord_flip() +
  scale_y_continuous(labels = scales::percent,
                    expand = expansion(mult = c(0, 0.15))) +
  labs(
    title = "Past-Year Substance Use Prevalence (NSDUH)",
    x = NULL,
    y = "Proportion reporting use",
    caption = paste("Total sample size: N =", format(nrow(NSDUH_2024),
big.mark = ", ")))
  ) +
  theme_minimal(base_size = 13)
```

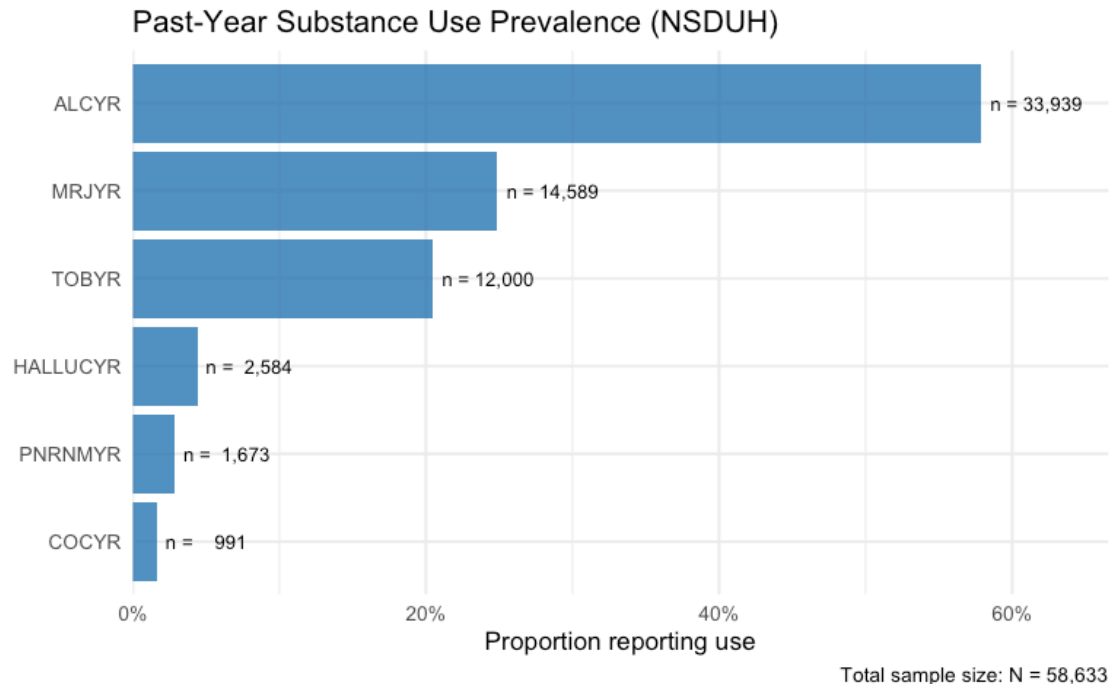


Figure 1. Past-Year Substance Use Prevalence (NSDUH 2024)

Table 2 and Figure 1 reveal substantial variation in past-year substance use prevalence across the six indicators. Alcohol is by far the most commonly reported substance, with over half the sample (~58%) reporting past-year use, followed by marijuana/cannabis at approximately 25% and tobacco/nicotine products at ~21%. In contrast, hallucinogens (~4%), prescription pain reliever misuse (~3%), and cocaine (~2%) were reported by a much smaller proportion of respondents. This variation in marginal prevalence is important context for the LCA: because some substances are more rare, we would expect classes defined primarily by those substances to also be small. These marginal frequencies, however, tell us nothing about how the substances co-occur within individuals, for instance, a person who uses alcohol may or may not also use cannabis, tobacco, or other substances. To understand that co-occurrence structure, we turn to pairwise association measures.

Next, we examine one pairwise contingency table to illustrate the concept. We chose to examine the association between the two most frequent substances, alcohol and marijuana/cannabis:

```
# Contingency table example: Alcohol x Cannabis
tab_alc_mj <- table(
  Alcohol = factor(NSDUH_2024$ALCYR, levels = c(1, 0), labels = c("Yes
Alcohol", "No Alcohol")),
  Cannabis = factor(NSDUH_2024$MRJYR, levels = c(1, 0), labels = c("Yes
Cannabis", "No Cannabis"))
)
addmargins(tab_alc_mj) |>
```

```
kable(caption = "Table 3. Contingency Table Example: Alcohol x Cannabis  
Past-Year Use")
```

Table 3. Contingency Table Example: Alcohol x Cannabis Past-Year Use

	Yes Cannabis	No Cannabis	Sum
Yes Alcohol	12455	21484	33939
No Alcohol	2134	22560	24694
Sum	14589	44044	58633

```
# Mosaic plot with shading for residuals
```

```
vcd::mosaic(tab_alc_mj,  
  shade      = TRUE,  
  legend     = TRUE,  
  main       = "Alcohol x Cannabis Co-occurrence",  
  labeling   = labeling_border(rot_labels = c(0, 0, 0, 0)))
```

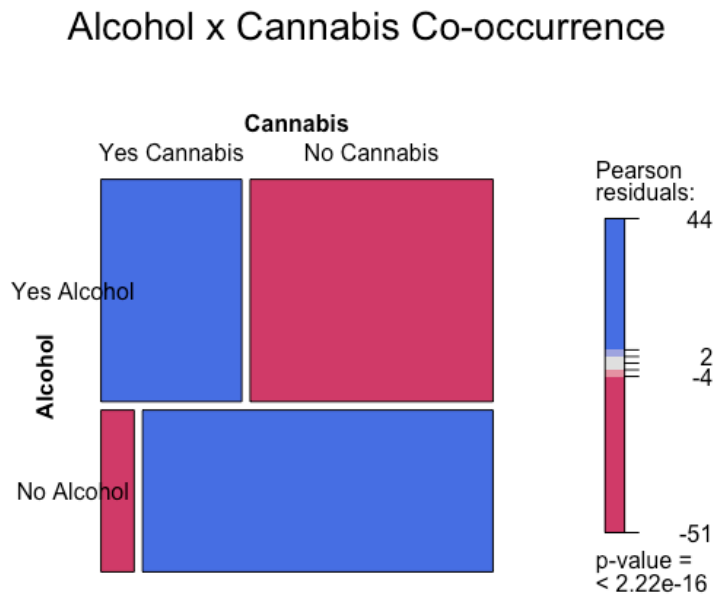


Figure 2. Mosaic Plot for the Association Between Alcohol and Marijuana/Cannabis

```
# Chi-square test
```

```
chisq.test(tab_alc_mj)
```

```
##
```

```
## Pearson's Chi-squared test with Yates' continuity correction
```

```
##
```

```
## data: tab_alc_mj
```

```
## X-squared = 6018.3, df = 1, p-value < 2.2e-16
```

The contingency table in Table 3 cross-tabulates past-year alcohol use against past-year marijuana/cannabis use. Among those who reported past-year alcohol use ($n = 33,939$), approximately 37% also reported past-year marijuana/cannabis use ($12,455/33,939$), compared to only about 9% of non-drinkers ($2,134/24,694$). In other words, alcohol users are roughly four times more likely to also use cannabis than non-drinkers. The chi-square test confirms that these two substances are not independent of one another: people who use alcohol are significantly more likely to also report cannabis use than we would expect by chance ($\chi^2(1) = 6018.3$, $p < .001$). This means that we can confidently reject the hypothesis that alcohol and marijuana/cannabis use are independent of one another in this sample.

Figure 2 presents a mosaic plot of the alcohol and marijuana/cannabis co-occurrence, where the area of each tile is proportional to the number of respondents in that cell, and the shading reflects Pearson residuals indicating how much each cell deviates from what would be expected under independence. Blue tiles indicate cells where the observed count is higher than expected, and red tiles indicate cells where it is lower than expected.

The pattern in the mosaic is highly significant ($p < .001$). The Yes/Yes cell (top left) is bright blue, meaning that co-use of alcohol and marijuana/cannabis occurs far more often than we would expect if the two substances were independent. Conversely, the Yes Alcohol/No Cannabis cell (top right) is bright red, indicating that alcohol-only use is less common than expected. Among non-drinkers, the pattern reverses: the No Alcohol/No Cannabis cell (bottom right) is bright blue, suggesting that abstaining from both substances is more common than chance would predict, while the No Alcohol/Yes Cannabis cell (bottom left) is bright red, meaning marijuana/cannabis-only use without alcohol is relatively rare.

While this contingency table and mosaic plot clearly illustrate a strong pairwise association, it only captures the relationship between two substances at a time. With six substances, there are 15 unique pairings, which is far too many tables/figures to present and interpret individually in a meaningful way. Instead, we summarize all 15 pairwise associations more efficiently using odds ratios, which reduce each 2×2 contingency table to a single number representing the strength of association between each pair of substances.

```
# Odds ratios for all pairs
pairs <- combn(substance_names, 2, simplify = FALSE)

or_results <- map_dfr(pairs, function(pair) {
  tab <- table(NSDUH_2024[[pair[1]]], NSDUH_2024[[pair[2]]])
  or <- (tab[1,1] * tab[2,2]) / (tab[1,2] * tab[2,1])
  tibble(
    Substance_1 = pair[1],
    Substance_2 = pair[2],
    OR           = round(or, 2)
  )
}) |> arrange(desc(OR))
```

```
kable(or_results, caption = "Table 4. Odds Ratios for all Substance Pairs")
```

Table 4. Odds Ratios for all Substance Pairs

Substance_1	Substance_2	OR
COCYR	HALLUCYR	27.90
MRJYR	HALLUCYR	26.65
MRJYR	COCYR	19.76
TOBYR	COCYR	13.27
COCYR	PNRNMYR	10.68
ALCYR	COCYR	9.38
ALCYR	HALLUCYR	7.85
ALCYR	MRJYR	6.13
TOBYR	HALLUCYR	5.39
TOBYR	MRJYR	5.27
PNRNMYR	HALLUCYR	5.14
ALCYR	TOBYR	3.64
TOBYR	PNRNMYR	3.54
MRJYR	PNRNMYR	3.47
ALCYR	PNRNMYR	1.69

Table 4 presents the odds ratios for all 15 pairwise substance combinations, sorted from strongest to weakest association. Every odds ratio is greater than 1, indicating that all six substances are positively associated with one another: users of any one substance are consistently more likely than non-users to also report using each of the other substances.

The strongest associations are found among the less common substances: for instance, cocaine and hallucinogens show the highest odds ratio (OR = 27.90), followed closely by marijuana/cannabis and hallucinogens (OR = 26.65) and marijuana/cannabis and cocaine (OR = 19.76). This suggests that although these substances are relatively rare in the sample individually, their users tend to strongly overlap: someone who uses cocaine is nearly 28 times more likely to also use hallucinogens than someone who does not use cocaine. In contrast, the weakest association is between alcohol and prescription pain reliever misuse (OR = 1.69). Alcohol and tobacco/nicotine products show a more modest association (OR = 3.64) relative to the illicit substance pairs, consistent with the fact that alcohol and tobacco are some of the most prevalent substances in the sample.

Taken together, these findings reveal a complex and uneven pattern of pairwise associations across all six substances. This complexity (15 associations of varying magnitude) illustrates the limitations of a purely pairwise CDA approach: while odds ratios efficiently summarize each pair, they cannot tell us how these associations combine to

form distinct subgroups of users in the population. This motivates the central question of this tutorial: can we identify a number of latent classes that simultaneously account for the full multivariate pattern of substance use across all six indicators? We turn to Latent Class Analysis to answer this question.

3. Fitting LCA Models

Setting Up the Model Formula

poLCA requires a formula specifying the observed variables. All variables must be coded as integers starting from 1. Since the NSDUH variables are coded 0 = No and 1 = Yes, we must recode our variables by adding 1 to each value. To do this, we begin by placing our six variables into a vector to make it easier to call them all at once, rather than typing all of them every time.

```
lca_vars <- c("ALCYR", "TOBYR", "MRJYR", "COCYR", "PNRNMYR", "HALLUCYR")
```

Next we recode the variables, adding 1 to shift values of 0 to 1 and values of 1 to 2. We use `lapply(...)` to move through each of the six variables in our vector and apply the same function to each one. The `as.integer(x + 1)` function tells R to add a value of 1 to every value, so if the original is equal to 0 (No), it'll be changed to 1, and if the original is equal to 1 (Yes), it'll be changed to 2. Note that the `as.integer()` wrapper makes sure the result is stored as a whole number rather than a decimal.

```
# Reload clean data
load("NSDUH_2024.RData")

# Recode: add 1 to shift 0/1 to 1/2
# 0 (No) becomes 1, 1 (Yes) becomes 2
NSDUH_2024[lca_vars] <- lapply(NSDUH_2024[lca_vars], function(x) {
  as.integer(x + 1)
})

# Verify that the recoding worked - for example, alcohol should show 24,694
# as 1 (No) and 33,939 as 2 (Yes)
lapply(NSDUH_2024[lca_vars], table)

## $ALCYR
##
##      1      2
## 24694 33939
##
## $TOBYR
##
##      1      2
## 46633 12000
```

```
##
## $MRJYR
##
##      1      2
## 44044 14589
##
## $COCYR
##
##      1      2
## 57642   991
##
## $PNRMYR
##
##      1      2
## 56960  1673
##
## $HALLUCYR
##
##      1      2
## 56049  2584
```

Now that we've verified that our recoding worked, we are finally able to define the model formula for poLCA. The `cbind()` portion bundles the six substance use variables together as the outcomes to be modeled, and the `~ 1` portion specifies that there are no predictors, meaning that we are finding classes purely from the pattern of substance use responses themselves.

Define the formula

```
f <- cbind(ALCYR, TOBYR, MRJYR, COCYR, PNRMYR, HALLUCYR) ~ 1
```

Please note that it was not necessary to include any code to address missing data, as the NSDUH dataset is complete (meaning that it does not contain any missingness on our variables of interest) - this was verified using the code below.

Check for missing values on each substance variable

```
colSums(is.na(NSDUH_2024[lca_vars]))
```

```
##      ALCYR      TOBYR      MRJYR      COCYR      PNRMYR      HALLUCYR
##         0         0         0         0         0         0
```

Fitting Models with 1–6 Classes

We fit models with increasing numbers of latent classes and compared fits using the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). Because poLCA uses random starting values, we ran each model multiple times and retained the best solution.

There are several features of the below code that are worth explaining. We use `set.seed(2024)` to make the results reproducible — because poLCA uses random starting

values, setting a seed ensures that anyone who runs this code with this seed value will get the same results. `max_classes <- 6` means we will try models with 1, 2, 3, 4, 5, and 6 latent classes. This number should match the number of models you want to fit. `n_reps <- 10` means we run each model 10 times with different random starting values, and we will keep the best result from those 10 runs.

Regarding the helper function, this code creates a custom function called `best_lca` that was written to help us handle the multiple runs of each model. The `best <- NULL` code section simply creates an empty placeholder where the best fitting model found so far can be stored. Next, `for (i in seq_len(nrep))` repeats the following steps `nrep` times (10 times in our case). Then, `fit <- polCA(...)` fits one LCA model with the specified number of classes. The included arguments tell R to run for a maximum of 1000 iterations (`maxiter = 1000`), to stop running when changes become very small (`tol = 1e-6`), and to avoid printing output for every run (`verbose = FALSE`). The code section `if (is.null(best) || fit$llik > best$llik) best <- fit` checks whether the present run produced a better log-likelihood than the previous best - if it did, or if this is the first run, it replaces `best` with the current model. Finally, the “best” at the end returns the best model found across all the runs.

Then we have the model fitting loop: the `lapply(...)` section loops through `k = 1, 2, 3, 4, 5,` and 6 and runs the following for each value of `k`. `cat("Fitting", k, "class model...")` tells R to print a progress message to the console so you can see which model is currently being fitted. This is useful simply because it takes the code a while to run. The `best_lca(f, NSDUH_2024, nclass = k, nrep = n_reps)` section of code calls our helper function above to fit and return the best model for `k` classes. Finally, the results of all six models are stored together in a list called “models”. `names(models) <- paste0("k", 1:max_classes)` gives each model in the list a readable name, `k1` through `k6`, so we can refer to them easily later on. For example, `models[["k3"]]` would give us the 3-class model.

```
set.seed(2024)
max_classes <- 6
n_reps      <- 10

# Helper: fit polCA nrep times, return best run
best_lca <- function(formula, data, nclass, nrep = 10) {
  best <- NULL
  for (i in seq_len(nrep)) {
    fit <- polCA(formula, data, nclass = nclass,
                  maxiter = 1000, tol = 1e-6, verbose = FALSE)
    if (is.null(best) || fit$llik > best$llik) best <- fit
  }
  best
}

# Fit models for k = 1 to 6 classes
models <- lapply(1:max_classes, function(k) {
  cat("Fitting", k, "class model...\n")
```

```

best_lca(f, NSDUH_2024, nclass = k, nrep = n_reps)
})

## Fitting 1 class model...
## Fitting 2 class model...
## Fitting 3 class model...
## Fitting 4 class model...
## Fitting 5 class model...
## Fitting 6 class model...

names(models) <- paste0("k", 1:max_classes)

models[["k1"]]

## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $ALCYR
##           Pr(1) Pr(2)
## class 1: 0.4212 0.5788
##
## $TOBYR
##           Pr(1) Pr(2)
## class 1: 0.7953 0.2047
##
## $MRJYR
##           Pr(1) Pr(2)
## class 1: 0.7512 0.2488
##
## $COCYR
##           Pr(1) Pr(2)
## class 1: 0.9831 0.0169
##
## $PNRNMYR
##           Pr(1) Pr(2)
## class 1: 0.9715 0.0285
##
## $HALLUCYR
##           Pr(1) Pr(2)
## class 1: 0.9559 0.0441
##
## Estimated class population shares
## 1
##
## Predicted class memberships (by modal posterior prob.)
## 1
##
## =====
## Fit for 1 latent classes:
## =====

```

```

## number of observations: 58633
## number of estimated parameters: 6
## residual degrees of freedom: 57
## maximum log-likelihood: -125738.4
##
## AIC(1): 251488.8
## BIC(1): 251542.7
## G^2(1): 23796.74 (Likelihood ratio/deviance statistic)
## X^2(1): 296514.9 (Chi-square goodness of fit)
##

models[["k2"]]

## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $ALCYR
##           Pr(1) Pr(2)
## class 1: 0.5432 0.4568
## class 2: 0.0930 0.9070
##
## $TOBYR
##           Pr(1) Pr(2)
## class 1: 0.9060 0.0940
## class 2: 0.4978 0.5022
##
## $MRJYR
##           Pr(1) Pr(2)
## class 1: 0.9542 0.0458
## class 2: 0.2051 0.7949
##
## $COCYR
##           Pr(1) Pr(2)
## class 1: 1.0000 0.0000
## class 2: 0.9376 0.0624
##
## $PNRNMYR
##           Pr(1) Pr(2)
## class 1: 0.9864 0.0136
## class 2: 0.9312 0.0688
##
## $HALLUCYR
##           Pr(1) Pr(2)
## class 1: 0.9996 0.0004
## class 2: 0.8384 0.1616
##
## Estimated class population shares
## 0.729 0.271
##
## Predicted class memberships (by modal posterior prob.)

```

```

## 0.7624 0.2376
##
## =====
## Fit for 2 latent classes:
## =====
## number of observations: 58633
## number of estimated parameters: 13
## residual degrees of freedom: 50
## maximum log-likelihood: -115127.3
##
## AIC(2): 230280.6
## BIC(2): 230397.3
## G^2(2): 2574.506 (Likelihood ratio/deviance statistic)
## X^2(2): 4431.556 (Chi-square goodness of fit)
##
models[["k3"]]

## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $ALCYR
##           Pr(1) Pr(2)
## class 1: 0.0681 0.9319
## class 2: 0.6324 0.3676
## class 3: 0.1759 0.8241
##
## $TOBYR
##           Pr(1) Pr(2)
## class 1: 0.3491 0.6509
## class 2: 0.9598 0.0402
## class 3: 0.6347 0.3653
##
## $MRJYR
##           Pr(1) Pr(2)
## class 1: 0.0595 0.9405
## class 2: 1.0000 0.0000
## class 3: 0.5112 0.4888
##
## $COCYR
##           Pr(1) Pr(2)
## class 1: 0.7700 0.2300
## class 2: 1.0000 0.0000
## class 3: 0.9963 0.0037
##
## $PNRNMYR
##           Pr(1) Pr(2)
## class 1: 0.8512 0.1488
## class 2: 0.9880 0.0120
## class 3: 0.9687 0.0313

```

```

##
## $HALLUCYR
##           Pr(1) Pr(2)
## class 1:  0.4545 0.5455
## class 2:  0.9992 0.0008
## class 3:  0.9819 0.0181
##
## Estimated class population shares
##  0.0673 0.5532 0.3795
##
## Predicted class memberships (by modal posterior prob.)
##  0.0516 0.6718 0.2766
##
## =====
## Fit for 3 latent classes:
## =====
## number of observations: 58633
## number of estimated parameters: 20
## residual degrees of freedom: 43
## maximum log-likelihood: -114082.7
##
## AIC(3): 228205.3
## BIC(3): 228384.9
## G^2(3): 485.252 (Likelihood ratio/deviance statistic)
## X^2(3): 714.8666 (Chi-square goodness of fit)
##
models[["k4"]]

## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $ALCYR
##           Pr(1) Pr(2)
## class 1:  0.5741 0.4259
## class 2:  0.1062 0.8938
## class 3:  0.0588 0.9412
## class 4:  0.2915 0.7085
##
## $TOBYR
##           Pr(1) Pr(2)
## class 1:  0.9952 0.0048
## class 2:  0.6615 0.3385
## class 3:  0.2203 0.7797
## class 4:  0.3107 0.6893
##
## $MRJYR
##           Pr(1) Pr(2)
## class 1:  0.9792 0.0208
## class 2:  0.1242 0.8758

```

```

## class 3: 0.0714 0.9286
## class 4: 0.7893 0.2107
##
## $COCYR
##           Pr(1) Pr(2)
## class 1: 1.0000 0.0000
## class 2: 1.0000 0.0000
## class 3: 0.6233 0.3767
## class 4: 0.9912 0.0088
##
## $PNRNMYS
##           Pr(1) Pr(2)
## class 1: 0.9874 0.0126
## class 2: 0.9656 0.0344
## class 3: 0.7986 0.2014
## class 4: 0.9610 0.0390
##
## $HALLUCYR
##           Pr(1) Pr(2)
## class 1: 0.9981 0.0019
## class 2: 0.8918 0.1082
## class 3: 0.4551 0.5449
## class 4: 1.0000 0.0000
##
## Estimated class population shares
## 0.6166 0.1889 0.0413 0.1532
##
## Predicted class memberships (by modal posterior prob.)
## 0.6519 0.2047 0.0363 0.1071
##
## =====
## Fit for 4 latent classes:
## =====
## number of observations: 58633
## number of estimated parameters: 27
## residual degrees of freedom: 36
## maximum log-likelihood: -113927.5
##
## AIC(4): 227908.9
## BIC(4): 228151.3
## G^2(4): 174.8369 (Likelihood ratio/deviance statistic)
## X^2(4): 244.8613 (Chi-square goodness of fit)
##
## ALERT: iterations finished, MAXIMUM LIKELIHOOD NOT FOUND
##
models[["k5"]]

## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)

```

```

##
## $ALCYR
##           Pr(1)  Pr(2)
## class 1:  0.4496 0.5504
## class 2:  0.1419 0.8581
## class 3:  0.9800 0.0200
## class 4:  0.2795 0.7205
## class 5:  0.0203 0.9797
##
## $TOBYR
##           Pr(1)  Pr(2)
## class 1:  0.1707 0.8293
## class 2:  0.6203 0.3797
## class 3:  1.0000 0.0000
## class 4:  0.8218 0.1782
## class 5:  0.2349 0.7651
##
## $MRJYR
##           Pr(1)  Pr(2)
## class 1:  0.5159 0.4841
## class 2:  0.0357 0.9643
## class 3:  0.9877 0.0123
## class 4:  0.9877 0.0123
## class 5:  0.0592 0.9408
##
## $COCYR
##           Pr(1)  Pr(2)
## class 1:  0.8438 0.1562
## class 2:  1.0000 0.0000
## class 3:  1.0000 0.0000
## class 4:  0.9993 0.0007
## class 5:  0.6084 0.3916
##
## $PNRNMYR
##           Pr(1)  Pr(2)
## class 1:  0.5940 0.4060
## class 2:  0.9694 0.0306
## class 3:  0.9873 0.0127
## class 4:  0.9854 0.0146
## class 5:  0.8206 0.1794
##
## $HALLUCYR
##           Pr(1)  Pr(2)
## class 1:  0.9224 0.0776
## class 2:  0.9084 0.0916
## class 3:  0.9999 0.0001
## class 4:  0.9942 0.0058
## class 5:  0.4251 0.5749
##
## Estimated class population shares

```

```

## 0.0128 0.2059 0.2534 0.4908 0.0371
##
## Predicted class memberships (by modal posterior prob.)
## 0.0064 0.2168 0.3568 0.3879 0.0322
##
## =====
## Fit for 5 latent classes:
## =====
## number of observations: 58633
## number of estimated parameters: 34
## residual degrees of freedom: 29
## maximum log-likelihood: -113885.1
##
## AIC(5): 227838.1
## BIC(5): 228143.4
## G^2(5): 90.07753 (Likelihood ratio/deviance statistic)
## X^2(5): 105.2961 (Chi-square goodness of fit)
##
## ALERT: iterations finished, MAXIMUM LIKELIHOOD NOT FOUND
##

models[["k6"]]

## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $ALCYR
##           Pr(1) Pr(2)
## class 1: 0.1184 0.8816
## class 2: 0.0270 0.9730
## class 3: 0.1964 0.8036
## class 4: 0.6228 0.3772
## class 5: 0.4948 0.5052
## class 6: 0.0326 0.9674
##
## $TOBYR
##           Pr(1) Pr(2)
## class 1: 0.8032 0.1968
## class 2: 0.2631 0.7369
## class 3: 0.5965 0.4035
## class 4: 0.9690 0.0310
## class 5: 0.1502 0.8498
## class 6: 0.1128 0.8872
##
## $MRJYR
##           Pr(1) Pr(2)
## class 1: 0.1519 0.8481
## class 2: 0.0781 0.9219
## class 3: 0.6150 0.3850
## class 4: 0.9955 0.0045

```

```

## class 5: 0.5405 0.4595
## class 6: 0.0274 0.9726
##
## $COCYR
##           Pr(1) Pr(2)
## class 1: 0.9997 0.0003
## class 2: 0.0259 0.9741
## class 3: 0.9964 0.0036
## class 4: 1.0000 0.0000
## class 5: 0.8551 0.1449
## class 6: 0.9107 0.0893
##
## $PNRNMYR
##           Pr(1) Pr(2)
## class 1: 0.9493 0.0507
## class 2: 0.7597 0.2403
## class 3: 0.9854 0.0146
## class 4: 0.9867 0.0133
## class 5: 0.5231 0.4769
## class 6: 0.8896 0.1104
##
## $HALLUCYR
##           Pr(1) Pr(2)
## class 1: 0.8086 0.1914
## class 2: 0.3376 0.6624
## class 3: 0.9985 0.0015
## class 4: 0.9988 0.0012
## class 5: 0.9066 0.0934
## class 6: 0.6187 0.3813
##
## Estimated class population shares
## 0.0957 0.0108 0.2816 0.5585 0.01 0.0433
##
## Predicted class memberships (by modal posterior prob.)
## 0.0218 0.0146 0.2853 0.6518 0.0059 0.0206
##
## =====
## Fit for 6 latent classes:
## =====
## number of observations: 58633
## number of estimated parameters: 41
## residual degrees of freedom: 22
## maximum log-likelihood: -113863.6
##
## AIC(6): 227809.3
## BIC(6): 228177.4
## G^2(6): 47.21539 (Likelihood ratio/deviance statistic)
## X^2(6): 51.15696 (Chi-square goodness of fit)
##

```

```
## ALERT: iterations finished, MAXIMUM LIKELIHOOD NOT FOUND
##
```

We can see in these results that there are alerts thrown for the 4-, 5-, and 6-Class solutions, meaning that the models did not fully converge even after 10 repetitions. This is strong evidence against using these solutions, regardless of what the AIC/BIC recommends. We turn to the next section to more fully evaluate our model fits and make a selection for our final model for interpretation.

4. Model Selection

For the model selection section of our tutorial, we've included `fit_stats <- map_dfr(1:max_classes, function(k) {` which loops through each of our fitted models ($k = 1$ through 6) and extracts fit statistics from each one, combining the results into a single data frame. `map_dfr` is like `lapply` but automatically stacks the results into rows of a table rather than a list. `m <- models[[k]]` means that for each value of k , R pulls out the corresponding model from our models list. So when $k = 3$, `m` becomes the 3-class model, for example. The `tibble()` block tells R to create one row of results for each model with 4 columns: `Classes` = the number of latent classes in the present model (eg., 1 through 6). `LogLik` = log-likelihood, which measures how well the model fits the data. A higher value means better fit, however, log-likelihood improves with the addition of more classes, so we can't use it as the only method from which a model is selected. `AIC` = the Akaike Information Criterion balances model fit against complexity by penalizing models with more parameters; the lower the AIC value the better. Lastly, `BIC` = the Bayesian Information Criterion is similar to AIC but penalizes complexity more heavily, which tends to favor simpler models. Again, the lower the BIC value, the better. The next chunk of code was included to add an asterisk to highlight the statistically best-fitting model. The last section of code is what generates the table. `kable()` renders the data frame as a formatted table.

Fit Statistics Table

```
fit_stats <- map_dfr(1:max_classes, function(k) {
  m <- models[[k]]
  tibble(
    Classes = k,
    LogLik  = round(m$loglik, 1),
    AIC     = round(m$aic, 1),
    BIC     = round(m$bic, 1),
  )
})

fit_stats |>
  mutate(
    BIC = ifelse(
      BIC == min(BIC),
```

```

    paste0(BIC, " *"),
    as.character(BIC)
  )
) |>
kable(
  caption = "Table 5. Model Fit Statistics for LCA Models with 1-6 Classes
(* = Lowest BIC)",
  col.names = c("Classes", "Log Likelihood", "AIC", "BIC"),
  align = "c"
)

```

Table 5. Model Fit Statistics for LCA Models with 1–6 Classes (= Lowest BIC)*

Classes	Log Likelihood	AIC	BIC
1	-125738.4	251488.8	251542.7
2	-115127.3	230280.6	230397.3
3	-114082.7	228205.3	228384.9
4	-113927.5	227908.9	228151.3
5	-113885.1	227838.1	228143.4 *
6	-113863.6	227809.3	228177.4

In our case, Table 5 shows that the 5-Class Solution is the statistically best-fitting model. But before we settle on this, we should consider the size of the improvements being made. From 1 Class to 2 Classes, the BIC reduced by 21,145.4 (a major improvement). Moving from 2 to 3 Classes showed a BIC improvement of 2,012.4, and moving from 3 to 4 Classes showed an improvement of 233.6. Moving from 4 to 5 Classes improved BIC by only 7.9 points, and we can see that fit got worse when going from 5 to 6 Classes (+33.0). These findings suggest that, while the 5 class model may statistically be the best fit, the low rates of improvement beyond 3 Classes suggest that the 3-Class Solution might actually be the better choice. This is especially true when we consider that the 4-, 5-, and 6-Class Solutions failed to converge on a stable maximum likelihood solution. As a result, these latter 3 models cannot be considered reliable, and model selections must prioritize solutions that converged properly. All this being said, it is advisable (and defensible) to choose the 3-Class Solution as our final model; it's nearly as good as the 4-Class Solution and will likely be easier to interpret meaningfully.

We also included/examined an elbow plot. The elbow plot shows visually what we learned through the above Table 5. The plot shows how AIC and BIC changed as we added more latent classes. Both lines drop sharply from 1 to 2 Classes, then again from 2 to 3 Classes, flattening out considerably from 3 Classes onwards. These plots are called elbow plots because you must look for the elbow - the point where adding another class yields diminishing improvement. The lowest BIC suggests the optimal number of classes was 5; however, substantive interpretability is equally important. Here, the elbow appears to be around 3 Classes. After that point, adding more classes gives diminishing returns in terms of model fit. Since the 4-, 5-, and 6-Class Solutions did not converge, I'm finally inclined to

go with the 3-Class Solution. To be sure, however, we will fit the 2-, 3-, 4-, and 5-class solutions and examine the item-response probability plots for each, so we can choose whichever model that produces classes that are more distinct and interpretable.

Elbow Plot

```
fit_stats_long <- fit_stats |>
  dplyr::select(Classes, AIC, BIC) |>
  pivot_longer(c(AIC, BIC), names_to = "Criterion", values_to = "Value")

ggplot(fit_stats_long, aes(x = Classes, y = Value, colour = Criterion)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  scale_x_continuous(breaks = 1:max_classes) +
  labs(
    title = "LCA Model Selection: AIC and BIC by Number of Classes",
    x = "Number of Latent Classes",
    y = "Information Criterion (lower = better)",
    colour = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "top")
```

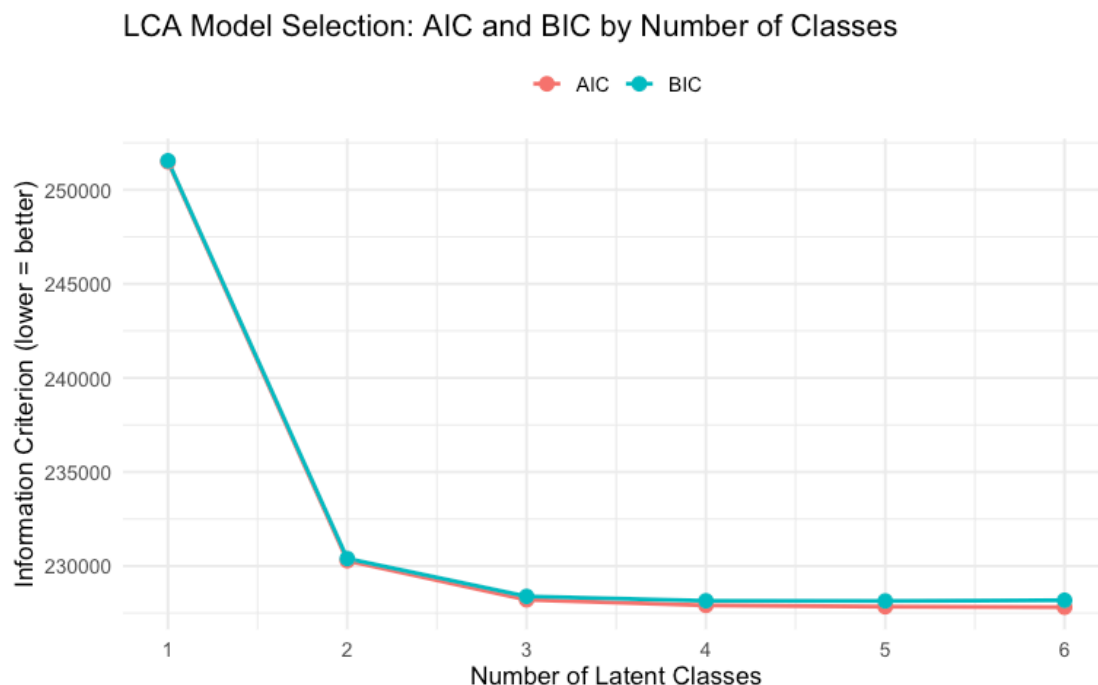


Figure 3. Elbow Plot for LCA Model Selection: AIC and BIC by Number of Classes

5. Interpreting the Best-Fitting Model

The heart of LCA interpretation lies in the item-response probabilities (IRPs): for each class and each substance, we are asking “what is the probability of “Yes” (past-year use)“?

We got the IRP matrices for the 1-, 2-, 3-, 4-, 5-, and 6-Class Solutions when we ran the models above. From this output, we can examine the probability of “Yes” for each substance/class. The 1- and 2-Class Solutions are coarse and lack detail, and we know from Table 5 and Figure 3 that the fit indices improved dramatically from the 1-Class Solution to the 2-Class Solution to the 3-Class Solution. We also know that the 4-, 5-, and 6-Class Solutions should not be considered given their non-convergence. Therefore, we focus our attention on the 3-Class Solution.

The 3-Class Solution identified three distinct substance use profiles. The first class (6.7%) was very likely to use alcohol, cannabis/marijuana, and tobacco, with moderate risk for hallucinogen use and low-to-moderate risk for cocaine and prescription drug misuse. This first class represents a higher-risk polysubstance use group. The second class in this solution (55.3%) was characterized by minimal substance use, with near-zero probabilities of illicit drug use, low tobacco use, and only moderate alcohol use, representing a near-abstinent group. The third class (38.0%) was very likely to use alcohol and was moderately likely to use tobacco and cannabis/marijuana, with little risk of using illicit drugs (cocaine, prescription drugs, and hallucinogens). This is a common developmental profile of social/recreational users who have not (yet) transitioned to broader polysubstance use patterns. This third class represents a licit substance use group.

Looking at the graph of IRPs below, the 3-Class Solution appears clean and interpretable (and it properly converged, unlike the 4-, 5-, and 6-class solutions). In the 3-Class LCA Solution, Class 1 is our Polysubstance Use Class, with elevated probabilities of using each of the six substances, Class 2 is a Low/No Use group, scoring moderately on alcohol and negligibly on tobacco, and Class 3 has high probabilities of using all the licit substances, such as alcohol, cannabis, and tobacco. This plot provides a visual to accompany the IRPs provided in the output above.

Item-Response Probability Plot

```
substance_labels <- c(
  ALCYR   = "Alcohol",
  TOBYR   = "Tobacco/Nicotine",
  MRJYR   = "Cannabis/Marijuana",
  COCYR   = "Cocaine",
  PNRNMYR = "Prescription Misuse",
  HALLUCYR = "Hallucinogens"
)
plot_irp <- function(model, k, substance_labels) {

  # Extract item-response probabilities
  irp <- map_dfr(seq_along(lca_vars), function(j) {
```

```

mat <- model$probs[[j]] # K x 2 matrix
tibble(
  Substance = substance_labels[lca_vars[j]],
  Class      = paste0("Class ", 1:nrow(mat)),
  P_Yes      = mat[, 2] # column 2 = P(category 2 = "Yes")
)
})

# Plot
ggplot(irp, aes(x = Substance, y = P_Yes, fill = Class)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = paste0(k, "-Class LCA Solution: Item-Response Probabilities"),
    x      = NULL,
    y      = "P(Past-Year Use | Class)",
    fill   = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "top",
        axis.text.x = element_text(angle = 20, hjust = 1))
}

# Plot 3-Class solution (5 is statistically best [lowest BIC], but since 4-,
# 5-, and 6-class solutions did not converge, the 3-class solution is likely to
# be a more interpretable/parsimonious option)
plot_irp(models[[3]], k = 3, substance_labels)

```

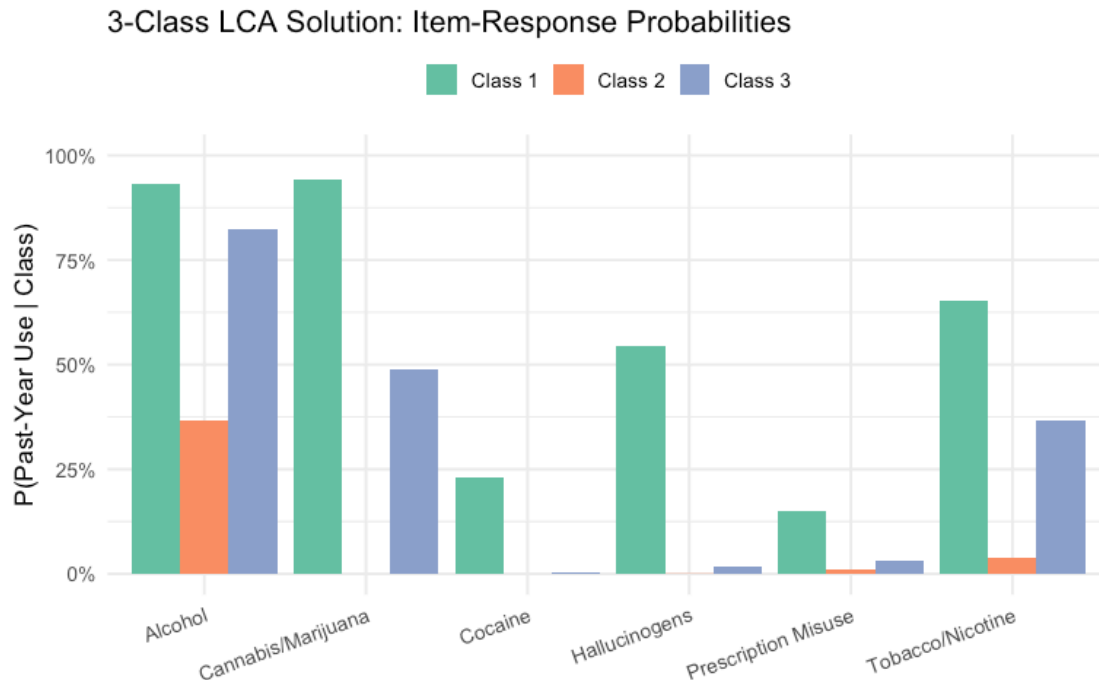


Figure 4. Bar Plot for LCA Model Selection: 3-Class Solution

6. Profiling the Classes

Finally, we examine how membership in the latent classes relates to demographic characteristics, assigning each respondent to their most probable class. The code below sets the number of classes (3) and extracts the corresponding LCA model from a list of fitted models (models). This ensures that “best_model” now contains the full 3-class solution. Next, we create a new variable that assigns each respondent to a latent class. Lastly, we label the classes as Class 1, Class 2, and Class 3 for ease of interpretation and for use in summaries and visualizations.

```
best_k <- 3
best_model <- models[[best_k]]

NSDUH_2024$predicted_class <- best_model$predclass
NSDUH_2024$predicted_class <- factor(NSDUH_2024$predicted_class,
                                     labels = paste0("Class ", 1:best_k))
```

Age Group Distribution by Class

```
# NSDUH age categories
age_labels <- c("12-17", "18-25", "26-34", "35+")
NSDUH_2024$age_group <- factor(NSDUH_2024$CATAGE, levels = 1:4, labels =
age_labels)
```

```
age_tab <- table(Age = NSDUH_2024$age_group, Class =
NSDUH_2024$predicted_class)
age_tab_prop <- prop.table(age_tab, margin = 2) # column proportions

# Mosaic plot: age x class
vcd::mosaic(age_tab,
  shade = TRUE,
  legend = TRUE,
  main = "Age Group Distribution by Latent Class",
  labeling = labeling_border(rot_labels = c(0, 0, 30, 0)))
```

Age Group Distribution by Latent Class

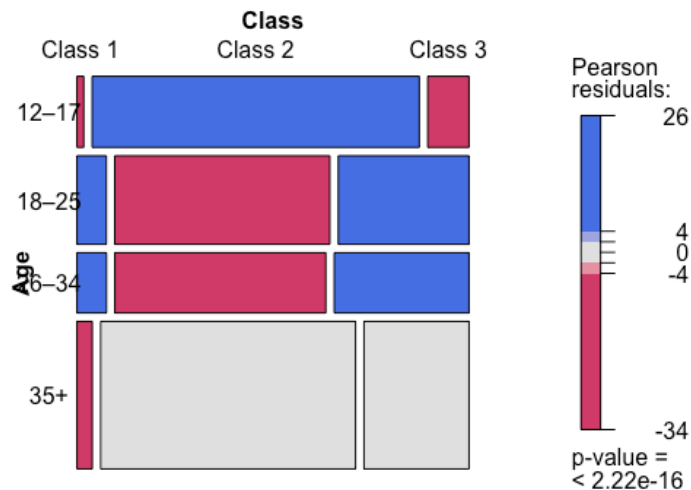


Figure 5. Mosaic Plot of Age x Class

Figure 5 shows that age distribution differed significantly across latent classes ($p < .001$). Class 1 - our polysubstance use class - is concentrated in young adults aged 18-34 years, with adolescents (12-17 years) and older adults (35+) being underrepresented. Class 2 - our near-abstinent class - is disproportionately composed of adolescents; young adults are strongly underrepresented. Finally, Class 3 - our licit substance use class - is overrepresented among young adults (18-34 years) and underrepresented among adolescents aged 12-17 years. There was no evidence for deviations from independence among older adults in either Class 2 or 3.

Sex Distribution by Class

```
sex_labels <- c("Male", "Female")
NSDUH_2024$sex <- factor(NSDUH_2024$IRSEX, levels = 1:2, labels = sex_labels)
sex_tab <- table(Sex = NSDUH_2024$sex, Class = NSDUH_2024$predicted_class)
```

```
vcd::mosaic(sex_tab,
  shade = TRUE,
  legend = TRUE,
  main = "Sex Distribution by Latent Class")
```

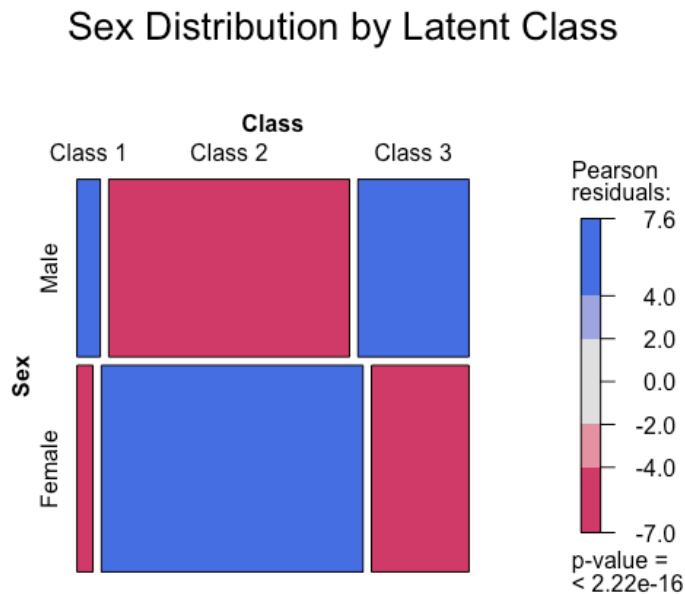


Figure 6. Mosaic Plot of Sex x Class

Figure 6 shows that sex distribution differed significantly across latent classes ($p < .001$). Class 1 - our polysubstance use class - is concentrated in males, with females being underrepresented. Class 2 - our near-abstinent class - is disproportionately composed of females; males are strongly underrepresented. Finally, Class 3 - our licit substance use class - is overrepresented among males and underrepresented among females.

Summary Table

```
profile_summary <- NSDUH_2024 |>
  group_by(predicted_class) |>
  summarise(
    N = n(),
    Pct_Male = scales::percent(mean(sex == "Male"), accuracy = 0.1),
    Pct_Youth = scales::percent(mean(age_group == "12-17"), accuracy = 0.1),
    Pct_EAs = scales::percent(mean(age_group == "18-25"), accuracy = 0.1),
    Pct_YAs = scales::percent(mean(age_group == "26-34"), accuracy = 0.1),
    Pct_OAs = scales::percent(mean(age_group == "35+"), accuracy = 0.1)
  )

kable(profile_summary,
  col.names = c("Class", "N", "% Male", "% Age 12-17", "% Age 18-25", "% Age 26-34", "% Age 35+"),
```

```
caption = "Table 6. Demographic Profile of Each Latent Class",
align = "c")
```

Table 6. Demographic Profile of Each Latent Class

Class	N	% Male	% Age 12-17	% Age 18-25	% Age 26-34	% Age 35+
Class 1	3026	55.6%	6.7%	36.5%	25.0%	31.7%
Class 2	39388	44.2%	25.1%	20.6%	13.7%	40.6%
Class 3	16219	49.5%	7.7%	30.4%	21.3%	40.6%

Table 6 presents the demographic composition of each latent class. We can see that Class 1 - our polysubstance use group - was skewed toward males (55.6%) and was concentrated among emerging (36.5%) and young (25.0%) adults, with relatively few adolescents (6.7%). In contrast, Class 2 - our near-abstinent group - was skewed toward females and was more broadly distributed across age but did indeed skew younger. Class 3 - our licit substance use group - showed a more balanced sex distribution (49.5% male) and skewed into adulthood (18+). Overall, these patterns indicate that latent class membership varies meaningfully across both age and sex, with distinct demographic profiles emerging for each class.

7. Discussion

In this tutorial, we demonstrated how Latent Class Analysis (LCA) can be used to identify distinct patterns of substance use in a large, multivariate categorical dataset (NSDUH 2024). Starting with six past-year substance use indicators, we first explored the data using classical categorical data analysis methods, including marginal frequencies, contingency tables, odds ratios, and mosaic plots. These exploratory analyses revealed strong pairwise associations between substances, particularly among less common illicit drugs, but also highlighted the limitations of purely pairwise approaches: as the number of variables increases, the number of possible combinations grows exponentially, making interpretation difficult.

By applying LCA, we summarized complex multivariate patterns with a small number of latent classes, each representing a profile of substance use behaviors. The iterative model-fitting process demonstrated key considerations in LCA:

- **Model Fit versus Convergence:** Although information criteria (AIC/BIC) initially suggested that larger models (5 classes) might fit better statistically, the 4-, 5-, and 6-class solutions failed to converge reliably. This illustrates an important practical point: statistical fit must be balanced with model stability and interpretability. Models that do not converge cannot be trusted, regardless of lower AIC or BIC values.

- **Elbow Criterion and Interpretability:** Visualizing fit indices with an elbow plot allowed us to see that improvements plateaued after the 3-class model. This highlights a common heuristic in LCA: the optimal number of classes is not always the one with the absolute lowest BIC, but rather the one that balances fit, parsimony, and meaningful interpretation.
- **Item-Response Probabilities (IRPs):** Examining the IRPs for the 3-class solution allowed us to label classes based on their substance use patterns. For example, we identified a polysubstance use class, a largely abstinent class, and a primarily alcohol/cannabis/tobacco use class. These distinctions illustrate how LCA can uncover subpopulations with qualitatively different risk profiles, which may not be apparent from univariate or pairwise analyses alone.
- **Demographic Profiling:** Classifying respondents into latent classes and cross-tabulating with demographic variables (age, sex) enables researchers to explore who is more likely to belong to each profile, providing actionable insights for prevention and intervention strategies.

Overall, this tutorial highlights the value of LCA as a tool for uncovering hidden subgroups in categorical data. Compared to traditional pairwise methods, LCA allows for a parsimonious and interpretable summary of complex co-occurrence patterns. At the same time, this exercise underscores the importance of careful model evaluation, including multiple random starts, convergence checks, and consideration of substantive interpretability alongside fit statistics.

For researchers and practitioners, the workflow presented here provides a practical blueprint:

- 1) Begin with exploratory analyses to understand marginal and pairwise patterns.
- 2) Fit LCA models with increasing numbers of classes and evaluate fit using AIC, BIC, and convergence diagnostics.
- 3) Examine IRPs and label classes meaningfully.
- 4) Profile classes by demographics or other relevant covariates.

By following this approach, researchers can confidently identify and interpret latent subgroups, enhancing both theoretical understanding and applied utility in substance use research. Future extensions of this work could include predictive modeling using latent classes, longitudinal LCA, or the incorporation of additional risk and protective factors to better understand trajectories of substance use across the lifespan.

References

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*This tutorial was developed as a course project for PSYC 6136: Categorical Data Analysis.

**Artificial Intelligence Disclosure: I used ChatGPT and Claude AI to help me solve errors in my R code, primarily when making the figures and tables, and to better understand how to interpret the latent classes. I tested all generated code with the NSDUH data set listed herein and the visual results made sense to me (and error messages disappeared), so I moved forward assuming that all was well. All written material reflects my own understandings/interpretations and conclusions.